A CRITERION REDUCING A PROPERTY OF COHERENT SHEAVES ON A NOETHERIAN ALGEBRAIC SPACE TO ITS BEHAVIOUR ON IRREDUCIBLE SUPPORTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Splitting a coherent sheaf along a union of two smaller closed supports. Same chapter spaces-cohomology.tex, label lemma-prepare-filter-support (invoked at proof line 2704). It produces the short exact sequence $0 \to \mathcal{G}_1 \to \mathcal{F} \to \mathcal{G}_2 \to 0$ with supports in the two strictly smaller closed pieces, which condition (1) turns into a contradiction.
- A power of the ideal of the support annihilates the sheaf. Same chapter spaces-cohomology.tex, label lemma-power-ideal-kills-sheaf (invoked at proof lines 2721 and 2762). It gives $\mathcal{J}^n\mathcal{F} = 0$ for the ideal defining the reduced structure on the support, and later it kills the kernel of the comparison map supported away from the open V.
- Preparation of a sheaf by powers of an ideal sheaf, via Artin-Rees. Same chapter spaces-cohomology.tex, labels lemma-prepare-filter-irreducible (invoked at proof line 2741) and lemma-Artin-Rees (invoked at proof line 2764). The first produces the injective maps $\mathcal{I}_1^{\oplus r_1} \to \mathcal{H}$ and $\mathcal{I}_2^{\oplus r_2} \to \mathcal{G}$ with cokernels supported in proper closed subsets and hence the open V over which $\mathcal{H}$ and $\mathcal{G}$ become isomorphic after taking powers; the second makes the resulting map injective after increasing the exponent.

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- Maps defined over an open extend after multiplying by a power of the ideal of the complement. Same chapter spaces-cohomology.tex, label lemma-homs-over-open (invoked at proof line 2755). It produces the map $\mathcal{I}^n\mathcal{G}^{\oplus r_1} \to \mathcal{H}^{\oplus r_2}$ which is an isomorphism over V.
- Coherent modules on a closed subspace are pushforwards of coherent modules on it. Morphisms of Spaces, label spaces-morphisms-lemma-i-star-equivalence (invoked at proof line 2738) and Same chapter spaces-cohomology.tex, label lemma-coherent-support-closed (invoked at proof line 2739). Together they express $\mathcal{F} = i_*\mathcal{H}$ with $\mathcal{H}$ coherent on the closed subspace Z defined by $\mathcal{J}$, which is what allows condition (3) to be applied.

Source reference. Original lemma 07UL: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2362–2362. Navigation pointer for \ref{lemma-Artin-Rees} used inside the counted span (line 2764); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UG: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2235–2235. Navigation pointer for \ref{lemma-coherent-support-closed} used inside the counted span (line 2739); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UM: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2389–2389. Navigation pointer for \ref{lemma-homs-over-open} used inside the counted span (line 2755); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UK: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2339–2339. Navigation pointer for \ref{lemma-power-ideal-kills-sheaf} used inside the counted span (line 2721); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UQ: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2478–2478. Navigation pointer for \ref{lemma-prepare-filter-irreducible} used inside the counted span (line 2741); the argument that consumes it is retained in the span.

Source reference. Original lemma 07UP: sources/stacks-spaces-cohomology/pinned-original-source.tex, source lines 2435–2435. Navigation pointer for \ref{lemma-prepare-filter-support} used inside the counted span (line 2704); the argument that consumes it is retained in the span.

Source reference. Original lemma 04CJ: sources/stacks-chow/tags.txt, source lines 5693–5693. Navigation pointer for \ref{spaces-morphisms-lemma-i-star-equivalence} used inside the counted span (line 2738); the argument that consumes it is retained in the span.

Source reference. Original lemma 04ZF: sources/stacks-chow/tags.txt, source lines 6463–6463. Navigation pointer for \ref{spaces-properties-lemma-Noetherian-topology} used inside the counted span (line 2696); the argument that consumes it is retained in the span.

Source reference. Original lemma 03IQ: sources/stacks-chow/tags.txt, source lines 4612–4612. Navigation pointer for \ref{spaces-properties-lemma-reduced-closed-subspace} used inside the counted span (line 2720); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let X be a Noetherian algebraic space over S. Let $\mathcal{P}$ be a property of coherent sheaves on X. Assume*

(1) *For any short exact sequence of coherent sheaves*
$$0 \to \mathcal{F}_1 \to \mathcal{F} \to \mathcal{F}_2 \to 0$$
if $\mathcal{F}_i$, $i = 1, 2$ have property $\mathcal{P}$ then so does $\mathcal{F}$.
(2) *If $\mathcal{P}$ holds for $\mathcal{F}^{\oplus r}$ for some $r \geq 1$, then it holds for $\mathcal{F}$.*
(3) *For every reduced closed subspace $i : Z \to X$ with $|Z|$ irreducible there exists a coherent sheaf $\mathcal{G}$ on Z such that*
   (a) *$\mathit{Supp}(\mathcal{G}) = Z$,*
   (b) *for every nonzero quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_Z$ there exists a quasi-coherent subsheaf $\mathcal{G}' \subset \mathcal{I}\mathcal{G}$ such that $\mathit{Supp}(\mathcal{G}/\mathcal{G}')$ is proper closed in $|Z|$ and such that $\mathcal{P}$ holds for $i_*\mathcal{G}'$.*

Then property $\mathcal{P}$ holds for every coherent sheaf on X.

Proof. Consider the collection
$$\mathcal{T} = \left\{ \begin{array}{c} T \subset |X| \text{ nonempty closed such that there exists a coherent sheaf} \\ \mathcal{F} \text{ with } \mathrm{Supp}(\mathcal{F}) = T \text{ for which the lemma is wrong} \end{array} \right\}$$
We are trying to show that $\mathcal{T}$ is empty. If not, then because $|X|$ is Noetherian (Properties of Spaces, Lemma 04ZF) we can choose a minimal element $T \in \mathcal{T}$. This means that there exists a coherent sheaf $\mathcal{F}$ on X whose support is T and for which the lemma does not hold.

If T is not irreducible, then we can write $T = Z_1 \cup Z_2$ with Z_1, Z_2 closed and strictly smaller than T. Then we can apply Lemma 07UP to get a short exact sequence of coherent sheaves
$$0 \to \mathcal{G}_1 \to \mathcal{F} \to \mathcal{G}_2 \to 0$$
with $\mathrm{Supp}(\mathcal{G}_i) \subset Z_i$. By minimality of T each of $\mathcal{G}_i$ has $\mathcal{P}$. Hence $\mathcal{F}$ has property $\mathcal{P}$ by (1), a contradiction.

Suppose T is irreducible. Let $\mathcal{J}$ be the sheaf of ideals defining the reduced induced closed subspace structure on T, see Properties of Spaces, Lemma 03IQ. By Lemma 07UK we see there exists an $n \geq 0$ such that $\mathcal{J}^n\mathcal{F} = 0$. Hence we obtain a filtration
$$0 = \mathcal{J}^n\mathcal{F} \subset \mathcal{J}^{n-1}\mathcal{F} \subset \ldots \subset \mathcal{J}\mathcal{F} \subset \mathcal{F}$$
each of whose successive subquotients is annihilated by $\mathcal{J}$. Hence if each of these subquotients has a filtration as in the statement of the lemma then also $\mathcal{F}$ does by (1). In other words we may assume that $\mathcal{J}$ does annihilate $\mathcal{F}$.

Assume T is irreducible and $\mathcal{J}\mathcal{F} = 0$ where $\mathcal{J}$ is as above. Denote $i : Z \to X$ the closed subspace corresponding to $\mathcal{J}$. Then $\mathcal{F} = i_*\mathcal{H}$ for some coherent $\mathcal{O}_Z$-module $\mathcal{H}$, see Morphisms of Spaces, Lemma 04CJ and Lemma 07UG. Let $\mathcal{G}$ be the coherent sheaf on Z satisfying (3)(a) and (3)(b). We apply Lemma 07UQ to get injective maps
$$\mathcal{I}_1^{\oplus r_1} \to \mathcal{H} \quad \text{and} \quad \mathcal{I}_2^{\oplus r_2} \to \mathcal{G}$$
where the support of the cokernels are proper closed in Z. Hence we find an nonempty open $V \subset Z$ such that
$$\mathcal{H}_V^{\oplus r_2} \cong \mathcal{G}_V^{\oplus r_1}$$
Let $\mathcal{I} \subset \mathcal{O}_Z$ be a quasi-coherent ideal sheaf cutting out $Z \setminus V$ we obtain (Lemma 07UM) a map
$$\mathcal{I}^n\mathcal{G}^{\oplus r_1} \longrightarrow \mathcal{H}^{\oplus r_2}$$
which is an isomorphism over V. The kernel is supported on $Z \setminus V$ hence annihilated by some power of $\mathcal{I}$, see Lemma 07UK. Thus after increasing n we may assume the

displayed map is injective, see Lemma 07UL. Applying (3)(b) we find $\mathcal{G}' \subset \mathcal{I}^n\mathcal{G}$ such that

$$(i_*\mathcal{G}')^{\oplus r_1} \longrightarrow i_*\mathcal{H}^{\oplus r_2} = \mathcal{F}^{\oplus r_2}$$

is injective with cokernel supported in a proper closed subset of Z and such that property $\mathcal{P}$ holds for $i_*\mathcal{G}'$. By (1) property $\mathcal{P}$ holds for $(i_*\mathcal{G}')^{\oplus r_1}$. By (1) and minimality of $T = |Z|$ property $\mathcal{P}$ holds for $\mathcal{F}^{\oplus r_2}$. And finally by (2) property $\mathcal{P}$ holds for $\mathcal{F}$ which is the desired contradiction. $\square$